Explore the storage options in AKS

- **Azure Disks**
    - Block-level storage volumes
    - Attached to Azure VMs, like AKS nodes, similar to how a real disk is attached to a server
    - Ideal for storing data that requires high performance and low latency
    - Mounted as ReadWriteOnce in AKS, so they can be attached only to one node

- **Azure Files**
    - Fully managed file shares in cloud using SMB, NFS and Azure Files REST APIs
    - Files can be shared and stored across multiple nodes
    - Needs connectivity on port 445 for SMB and on 111/2048 for NFS for the file endpoint

- **Azure Blob**
    - Optimized for storing massive amounts of unstructured data
    - Intended for images, documents, and audio or video streaming, logging data to log files, disaster recovery data and others
    - Uses BlobFuse, for which requires connectivity on port 443 and NFS for which requires connectivity on 111/2048 for the blob endpoint

- **Others:**
    - standalone NFS server, Azure NetApp Files, Azure Ultra Disks

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- CSI driver is used to control the lifecycle of Azure storage resources and helps with new features

- Using CSI drivers in AKS avoids making changes to the main Kubernetes code or waiting for its release cycles

- **CSI added features**
  - **Azure Disk**
    - Better attach and detach performance
    - Premium SSD v1 and v2 are supports
    - Zone-redundant storage (ZRS) disk support
    - Snapshot
    - Volume clone
    - Resize disk PV without interrupting operations

  - **Azure File**
    - Can use NFS v4.1
    - Private endpoint
    - Create large mounts of file shares in parallel

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- Provisioning options
  - **Dynamic**
    - Relies on Storage Classes
      - Defines various storage types, enabling the user to request a particular type of storage for their workloads
      - AKS comes with pre-created Storage Classes
      - You can create custom Storage Classes
      - The Storage Class needs to be referenced at PVC level
      - Azure Disk specific: You will need to reference the associated PVC to a pod to provision the disk
      - Azure File and Blob specific: A secret containing the Storage Account credentials is automatically created

  - **Static**
    - You create the Azure Disk or Storage Account (for Azure File or Blob) by yourself
    - Azure Disk specific: You will create a PV and PVC
    - Azure File or Blob specific: You will also need to create the secret manually
      - You can mount as:
        - Inline volume
        - Create PV and PVC